The Immunometabolic Architect: A Research Dossier on Gökhan Sıddık Hotamışlıgil

A) ANCHOR SCENES

SCENE #1: The 1993 Paradox: Transforming a Wasting Factor into a Metabolic Driver Summary: This scene marks the foundational moment of the field of immunometabolism, occurring when Gökhan Hotamışlıgil, then a postdoctoral fellow, demonstrated that Tumor Necrosis

Factor-alpha (TNF-α), a cytokine previously understood exclusively as an immune system signaling molecule associated with "cachexia" or wasting, was actively expressed in the adipose tissue of obese organisms and served as a primary driver of insulin resistance. Date/Place: Early 1993; The laboratory of Bruce M. Spiegelman, Dana-Farber Cancer Institute and Harvard Medical School, Boston, Massachusetts. [1] People: Gökhan S. Hotamışlıgil, Bruce M. Spiegelman, and Narinder S. Shargill. [4] Concrete details: The research environment was defined by the high-stakes molecular biology culture of the Dana-Farber Cancer Institute, where the Spiegelman lab focused on the transcriptional control of adipocyte differentiation.

Hotamışlıgil utilized Northern blot analysis to quantify the expression of TNF-α mRNA across a spectrum of rodent models, including the leptin-deficient ob/ob mouse, the leptin-receptor-deficient db/db mouse, the fa/fa Zucker rat, and mice with obesity induced by monosodium glutamate (MSG). [2] The experimental texture involved the painstaking isolation of mRNA from white adipose tissue and the observation of a significant 2.5-fold increase in TNF-α expression in obese subjects compared to lean controls. [3] The "pivot point" occurred when the team neutralized TNF-α in the fa/fa rat model using a soluble TNF-receptor-IgG fusion protein. The resulting data showed a dramatic improvement in peripheral glucose uptake, effectively proving that the cytokine was not just present but was functionally blocking the metabolic action of insulin. [2] Why it matters: This discovery effectively dismantled the prevailing "passive storage" model of adipose tissue. It suggested that obesity was not merely a state of excess caloric storage but a state of chronic, low-grade inflammation. This provided the first mechanistic bridge between the immune system and metabolic homeostasis, laying the groundwork for the modern understanding of type 2 diabetes as an inflammatory disease. [8] Evidence quality: Primary. The foundational research was published in *Science* in 1993 ($Science$, 259, 87-91) and reinforced by clinical human data in *The Journal of Clinical Investigation* in 1995. [2] Myth risk: Medium. Popular scientific retelling often frames this as a singular "Eureka" moment in front of a film developer, while the

data indicates a systematic, multi-model validation process spanning several years of postdoctoral work. [1] Sources: [1]

SCENE #2: The 2002 JNK Smoking Gun: Mapping the Molecular Sabotage Summary: Building on the earlier inflammatory discovery, this scene documents the lab's identification of the c-Jun N-terminal kinase (JNK) as the specific enzyme that intercepts the insulin signaling pathway, providing the definitive biochemical link between metabolic stress and cellular insulin resistance. Date/Place: November 2002; Harvard School of Public Health, Department of Nutrition. [4] People: Gökhan S. Hotamışlıgil, Jiro Hirosumi (lead author), Gürol Tuncman, and Cem Z. Görgün. [12] Concrete details: The experimental work relied on the creation and analysis of $JNK1$ -deficient mice. The laboratory setting involved complex metabolic flux measurements and insulin clamp techniques. [12] Hotamışlıgil's team observed that JNK activity was abnormally elevated in the liver, muscle, and adipose tissues of obese mice. [12] The critical biochemical evidence was the discovery that JNK phosphorylates the insulin receptor substrate-1 (IRS-1) at the Serine 307 residue. In the context of intro biochemistry, this is a quintessential example of inhibitory phosphorylation: while tyrosine phosphorylation is required for insulin signaling, JNK-mediated serine phosphorylation acts as a molecular "off-switch." [4] Why it matters: This was the first time a specific kinase was identified as a targetable "node" connecting inflammatory signals (like TNF-α) to metabolic failure. It validated the entire concept of "metaflammation" by showing exactly how a cell "decides" to stop responding to insulin when it is under metabolic stress. [4] Evidence quality: Primary. Published in *Nature* in 2002 ($Nature$, 420, 333-6). [4] Myth risk: Low to Medium. While the discovery is foundational, a 2023 author correction was issued regarding specific blot images in the original paper, though the core scientific conclusions regarding JNK and IRS-1 remain widely accepted in the field. [13] Sources: [4]

SCENE #3: The 1980 Ankara Coup: Scientific Ambition Amidst Political Collapse Summary: This scene explores the formative years of Hotamışlıgil's medical training, which occurred during the 1980 military coup in Turkey. It highlights the institutional barriers and the role of serendipity in his eventual move to the United States. Date/Place: September 1980 – 1986; Ankara University Medical School, Ankara, Turkey. [1] People: Gökhan S. Hotamışlıgil (Medical Student), and his wife. [1] Concrete details: Hotamışlıgil entered medical school during a period of extreme civil unrest and political violence, which culminated in the September 12, 1980, military coup. [1] The university environment was highly politicized, with frequent disruptions. He specialized in pediatrics and completed a residency at Ankara University. [1] The pivotal "career redirection" occurred when his wife was awarded a scholarship to the Eunice Kennedy Shriver Center for Mental Retardation in Massachusetts. [1] Faced with the choice of remaining in Turkey to fulfill compulsory government medical service in the remote town of Bingöl or finding a way to join his wife in the U.S., Hotamışlıgil began cold-calling Harvard laboratories. [1] Why it matters: This scene illustrates the transition from a clinical medical path in an emerging

economy to a basic research path in the world's leading scientific hub. It also highlights his resilience in navigating institutional barriers, as he transitioned from neuro-genetics (studying monoamine oxidases in Xandra Breakefield's lab) to molecular metabolism. [1] Evidence quality: Primary. Detailed oral history provided by the scientist to the Science History Institute. [1] Myth risk: Low. These biographical details are self-reported in an academic context and consistent with the historical record of the 1980 coup. [1] Sources: [1]

SCENE #4: Mapping the "Melting" Liver: 3D Visualization of Organelle Decay (2022) Summary: In a move from molecular signaling to structural biology, the Hotamışlıgil lab utilized advanced electron microscopy to visualize the liver cell's physical architecture, demonstrating that obesity causes a physical "unraveling" of the connections between the endoplasmic reticulum (ER) and mitochondria. Date/Place: March 2022; Sabri Ülker Center for Metabolic Research, Harvard T.H. Chan School of Public Health, Boston. [14] People: Gökhan S. Hotamışlıgil and Gunes Parlakgül. [13] Concrete details: The team employed Focused Ion Beam Scanning Advanced Electron Microscopy (FIB-SEM). This technology allowed them to generate 3D reconstructions of liver cells at an unprecedented resolution. [14] They compared "lean" liver tissue with "obese" tissue, revealing that in obesity, the highly ordered, tubular structure of the ER becomes disorganized and fragmented, losing its physical proximity to the mitochondria. [14] This structural disarray was shown to be a primary cause of metabolic dysfunction. The scene concludes with a dramatic intervention: by molecularly "repairing" these structural contacts, they were able to restore normal metabolism in the liver. [14] Why it matters: This research shifted the paradigm from "signaling pathways" to "organelle architecture." It suggested that the physical geometry of the cell is just as important as the chemical signals it receives. It introduced the concept of "organelle homeostasis" as a core pillar of metabolic health. [9] Evidence quality: Primary. Published in *Nature* in 2022 ($Nature$, 603, 736–742). [14] Myth risk: Low. The technological specificity of FIB-SEM provides a strong evidentiary anchor. [14] Sources: [9]

SCENE #5: The 2025 Translational Leap: Launching the Enlila Venture Summary: This scene captures the lab’s evolution from basic discovery to therapeutic application, marked by a massive 10-year sponsorship from İşBank and the founding of a biotech company to bring "Fabkin" antibodies to the clinic. Date/Place: June 16, 2025; Sabri Ülker Center, Boston. [19] People: Gökhan S. Hotamışlıgil and executives from İşBank and İş Private Equity. [20] Concrete details: The event was a formal signing ceremony for a 10-year research agreement. [20] This agreement facilitated the launch of **Enlila**, a biotech venture aimed at translating the lab's research on secreted fatty acid-binding proteins (FABPs) and the hormone complex "Fabkin" into treatments for human disease. [19] The workspace at this stage is a mix of academic inquiry and high-level biotechnology partnership, focusing on monoclonal antibodies designed to neutralize circulating FABP4. [13] Why it matters: It represents the "full circle" of Hotamışlıgil's career: from identifying a single gene in 1993 to launching a multi-national therapeutic platform in 2025. It also underscores his continued influence on the Turkish scientific and

financial ecosystem. [1] Evidence quality: Primary. Official institutional announcements and lab history updates. [19] Myth risk: Low. This is a recent, well-documented institutional event. [20] Sources: [1]

Comparative Analysis of Rodent Models Used in Foundational Scenes

The following table details the specific animal models utilized in the initial discovery of the TNF-α metabolic link, demonstrating the breadth of the scientist's systematic approach.

Model Designation	Species	Genetic Basis	Metabolic Observation	Source
ob/ob	Mouse	Leptin Deficiency	Massive obesity; severe insulin resistance	[2]
db/db	Mouse	Leptin Receptor Deficiency	Severe obesity; type 2 diabetes phenotype	[2]
fa/fa	Rat	Leptin Receptor Mutation	Hyperphagia; early-onset insulin resistance	[2]
MSG-Induced	Mouse	Chemical/Hypothalamic Damage	Obesity without primary genetic mutation	[2]
Zucker	Rat	Spontaneous Genetic Mutation	Standard model for metabolic syndrome	[2]

B) DIRECT QUOTES FROM THE SCIENTIST

“Prof. Hotamışlıgil always reminds us that the most important thing is not money, or fame, or publishing, but to do the highest quality and true science so that we can understand more about the biology of living organisms and nature.” Context/Date: Attributed to Hotamışlıgil by his students, Drs. Güney and Parlakgül, during their fellowship in the lab (circa 2011-2014). [21] Source: Harvard T.H. Chan School of Public Health, Sabri Ülker Fellows Profile. [21]

“Gökhan Sıddık Hotamışlıgil is a quite long name and hard to pronounce. Sometimes they call me ‘Hotalar.’ But mostly Americans shorten Gökhan and call me ‘Dr. G’ and choose the easy way. Indeed, having a name that is difficult to pronounce abroad is problematic.” Context/Date: Interview with the Turkish newspaper *Yeni Akit*, October 8, 2014, regarding his identity as a Turkish scientist in the United States. [22] Source: Wikipedia (archived citation from *Yeni Akit*). [22]

“The discovery that obesity itself results in an inflammatory state in metabolic tissues ushered in a research field that examines the inflammatory mechanisms in obesity.” Context/Date: From a comprehensive review article titled "Inflammatory Mechanisms in Obesity" published in 2011, reflecting on the 18-year legacy of his 1993 discovery. [10] Source: *Annual Review of Immunology*, Vol. 29, 2011. [10]

“There were three people who gave me the most perspective during my formative years: my father taught me to try to be the best in everything I do, my mother taught me to relax and enjoy life through art and music, and my grandfather was an example for me about how to care for the wellbeing of people.” Context/Date: A personal reflection on his upbringing in Turkey provided during the inauguration of the Sabri Ülker Center. [21] Source: Sabri Ülker Foundation official records. [21]

“As we venture into this long and arduous path to reach our goals, we have many bright ideas, yet we realize that what is expected is their implementation, and what counts will be our deeds. The tasks belong to each and every one of us to be shouldered and carried out.” Context/Date: Keynote address at the First Annual Conference of the Turkish American Scientists and Scholars Association (TASSA), 2005. [18] Source: TASSA Conference Proceedings, 2005. [18]

“The Ottoman case is a striking example of what happens to a country if scientific research is neglected for many years... The Ottoman scholars did not try to participate in developing the new fields opened up by scientific research, based on observation, experimentation and mathematical formulation of theories.” Context/Date: A historical and philosophical lecture on the state of Turkish science, delivered at the TASSA conference in Washington D.C., 2005. [18] Source: TASSA Conference Proceedings, 2005. [18]

C) HUMAN REALITY DETAILS

Hotamışlıgil was born in Pazar, a small town on the Black Sea coast of Turkey, but his

childhood was nomadic; he attended three different elementary schools as his father, a physician, moved the family to provide medical care in underdeveloped and underserved regions of Turkey. [1] This early exposure to rural clinical practice significantly shaped his later focus on public health and metabolic disease.

Despite the political instability of the 1980s, he was a varsity athlete in high school, competing in track and field, volleyball, and table tennis, which he credits with instilling a sense of competitive discipline and teamwork. [1]

His entry into the world of molecular research was largely accidental; his interest in genetics was sparked not by his medical classes, which were largely theoretical, but by a seminar he prepared on thalassemia, a common Mediterranean blood disorder. [1]

The decision to move to the United States was a matter of marital logistics; he sought a research position at Harvard specifically because his wife had received a scholarship to the Shriver Center in Massachusetts, and he wanted the family to remain together. [1]

In the laboratory, he is known for a mentoring style that emphasizes the "truth of biology" over the pressures of the "publish or perish" culture, a philosophy that has helped him train over 90 scientists who now hold independent positions globally. [20]

His lab culture is uniquely international and multidisciplinary, frequently recruiting Turkish scientists through the Sabri Ülker Fellowship program, which aims to bridge the gap between Turkish medical education and American basic research. [14]

He maintains a balance between high-intensity research and personal well-being by following his mother's advice to engage with art and music, a trait that colleagues say prevents the "burnout" common in elite academic departments. [21]

His colleagues and students affectionately refer to him as "Dr. G," a moniker that serves as both a pragmatic solution to his difficult-to-pronounce surname and a sign of the approachable, less-hierarchical atmosphere he fosters in his lab. [22]

He serves on the Board of Trustees for Kadir Has University in Turkey, indicating a career-long commitment to institutionalizing scientific research in his home country. [22]

During the 1980 coup, he faced the high-stakes pressure of taking medical entrance exams while military tanks were on the streets, an experience he describes as a formative lesson in focusing on goals despite external chaos. [1]

He is an Associate Member of several of the world's most prestigious research institutes, including the Broad Institute of MIT and Harvard, the Harvard Stem Cell Institute, and the Joslin Diabetes Center. [24]

His father's influence was paramount; he was taught that being a physician was not just a job but a social responsibility to "be the best in everything" for the sake of the patient. [21]

Key Career Milestones and Institutional Affiliations

Period	Role / Position	Institution	Primary Focus	Source
1980-1986	Medical Student	Ankara University	Clinical Medicine & Pediatrics	[1]
1988-1989	Research Fellow	Harvard Medical School	Neuro-genetics (Breakefield Lab)	[1]
1990-1994	PhD Candidate	Harvard Medical School	Biological Chemistry (Spiegelman Lab)	[1]
1995-Present	Professor / Chair	Harvard T.H. Chan School of Public Health	Molecular Metabolism & Genetics	[1]
2014-2024	Director	Sabri Ülker Center	Nutrient & Metabolic Research	[16]
2025-Present	Founder / Advisor	Enlila Biotech	Translational Therapeutics	[19]

D) SCIENCE CONNECTION (Intro Biochemistry Bridge)

Best-fit intro biochemistry topic: Metabolic Signaling and Post-Translational Modification.

Precise contribution: Hotamışlıgil demonstrated that obesity-induced inflammatory signals (TNF-α) activate the kinase JNK, which sabotages the insulin pathway by phosphorylating the Insulin Receptor Substrate-1 (IRS-1) at inhibitory serine residues instead of the activating

tyrosine residues. [4]

Teachable bridge explanation for students: In an introductory biochemistry course, students learn that the insulin receptor is a receptor tyrosine kinase. When insulin binds, the receptor phosphorylates itself and then phosphorylates a "docking" protein called IRS-1 on specific tyrosine (Y) residues. This tyrosine phosphorylation is the "green light" for the cell to transport glucose. Hotamışlıgil's research introduced the "red light" mechanism: in a state of obesity, the body produces inflammatory cytokines like TNF-α. These cytokines activate an enzyme called JNK (c-Jun N-terminal kinase). JNK "mis-phosphorylates" IRS-1 by adding a phosphate group to a Serine (S) residue (specifically Ser307 in mice). This serine phosphorylation changes the shape of IRS-1, preventing it from interacting with the insulin receptor. Consequently, even if insulin is present, the "green light" is blocked, leading to insulin resistance and type 2 diabetes. [4]

Scientist's own framing/analogy: Hotamışlıgil uses the analogy of a nation's "economic and defense systems." [25] The metabolic system (economy) and the immune system (defense) are traditionally seen as separate. However, in obesity, the "economic excess" (over-nutrition) is perceived by the body as a "defense threat." The immune system's response—inflammation—is like a military intervention that accidentally destroys the domestic economic infrastructure (the insulin signaling pathways), leading to a state of internal collapse (metabolic disease). [25]

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

Item: The Reproducibility and Image Integrity Scrutiny (PubPeer) What's agreed: Over the past two decades, several papers from the Hotamışlıgil lab have been flagged on PubPeer for irregularities, specifically duplicated or manipulated images in Western blots. [16] Formal corrections have been issued for a 2002 *Nature* paper, a 2008 *PLOS One* paper, and a 2020 *Cell Metabolism* paper. [17] What's disputed: Critics like computational biologist Reese Richardson have argued that these errors suggest systemic methodological inconsistencies or reporting errors in the lab's statistical analyses. [17] Hotamışlıgil has countered that many of these critiques are "without basis" or involve decades-old data for which raw blots are no longer available due to standard laboratory retention policies. [17] He maintains that the core scientific findings of these papers have been independently replicated by numerous other laboratories. [16] Best sources: *Retraction Watch* (July 2025 investigative report); PubPeer database; *Nature* Author Corrections (2023). [13]

Item: The Paradoxical Nature of Metaflammation What's agreed: Obesity results in a state of chronic, low-grade inflammation termed "metaflammation." [10] This state is generally considered detrimental as it leads to diabetes and heart disease. [23] What's disputed: A growing body of "emerging evidence" suggests that metaflammation may not be purely pathological. Some researchers, and increasingly the Hotamışlıgil lab itself, are exploring

whether the early inflammatory response in adipose tissue is actually an **adaptive** mechanism meant to help the body safely expand fat stores and maintain temporary homeostasis under nutrient stress. [28] The dispute lies in determining the exact "tipping point" where this defense mechanism transitions from being helpful to being harmful. [28] Best sources: Yun Sok Lee, "Immunometabolism in obesity: Understanding the beneficial and detrimental roles of inflammation," *PLoS Biol* (2026); Hotamışlıgil, "Inflammation, metaflammation and immunometabolic disorders," *Nature* (2017). [8]

Item: Credit Assignment in Collaborative Discoveries What's agreed: Hotamışlıgil has co-authored several high-profile papers with researchers who have faced significant scientific misconduct allegations, such as Michael Karin and Umut Ozcan. [17] What's disputed: In the world of high-impact biology, the senior author (Hotamışlıgil) is often the focus of both the praise and the scrutiny. The dispute often centers on the degree of oversight a senior PI can reasonably maintain over collaborative work performed in other labs or by independent postdocs. [17] Best sources: *Retraction Watch*; *Science* Magazine commentary. [17]

F) "NOT ON WIKIPEDIA" GOLD

- **The "Adipo-Pulmonary Axis" (2024):** Beyond diabetes, Hotamışlıgil's recent work has identified a specific hormonal link between the liver/fat and the lungs. His lab discovered that the protein FABP4 (aP2) acts as a hormone that can trigger airway inflammation, explaining why obese patients are significantly more likely to suffer from severe, treatment-resistant asthma. This "adipo-pulmonary axis" is a burgeoning sub-field not yet detailed on standard biographical pages. [13]
- **The Discovery of "Fabkin" (2021):** While Wikipedia mentions FABP4, it often misses the discovery of the **Fabkin** complex. Hotamışlıgil discovered that FABP4 does not act alone but forms a complex with two extracellular nucleoside kinases (ADK and NDPK). This complex (Fabkin) is what actually docks onto pancreatic beta cells to sabotage insulin production, representing a major leap in understanding both Type 1 and Type 2 diabetes. [14]
- **The "Hotalar" Moniker:** Hotamışlıgil reveals in Turkish interviews that his Turkish colleagues often call him "Hotalar" (a pluralized version of the start of his surname), a term of endearment that signifies his role as a "father figure" or leader of a large "scientific clan" of Turkish researchers in the diaspora. [22]
- **İşBank Partnership as a National Milestone:** The 2025 deal with İşBank is not just a research grant; it is the largest sponsored research agreement in the history of the Harvard Chan School with a Turkish institution, signifying a major shift in how international private equity is funding basic science at Harvard. [17]
- **The "Melted" ER Visualization:** The lab's recent use of FIB-SEM electron microscopy provided the first 3D "visual proof" of metabolic disease. Hotamışlıgil describes the obese liver cell's ER not just as "stressed" but as physically "unraveled" or "melted," a visceral structural description that moves beyond traditional molecular pathway

diagrams. [14]

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H) FLAGGED “MYTH-RISK” DETAILS

Claim: Hotamışlıgil coined the term "metaflammation" in his 1993 *Science* paper. **Where it appears:** Common in student presentations and some introductory blog posts on the "Western Diet." [27] **Why sourcing is weak:** A search of the original 1993 paper (*Science* 259,

87-91) does not contain the word "metaflammation." PubMed records indicate the term first began appearing in scientific literature around 2006-2008. [26] **Best available evidence:** Hotamışlıgil did indeed develop the *concept* in 1993, but the *terminology* was a later synthesis intended to differentiate "metabolic inflammation" from "classical immune inflammation." [10]

Claim: His discovery of JNK was an "overnight breakthrough." **Where it appears:** Early press releases from the ADA and various "Scientist of the Year" profiles. [30] **Why sourcing is weak:** The 2002 JNK paper lists a team of researchers and involves data from multiple mouse strains ($JNK1-/-, JNK2-/-$) and tissue types. [12] **Best available evidence:** The discovery was a three-year systematic genetic screening process. The 2023 correction of the paper's blot images further suggests that the "vividness" of the original data required significant post-hoc clerical adjustment. [17]

Claim: He fled Turkey as a political refugee after the 1980 coup. **Where it appears:** Occasionally mentioned in social media discussions regarding the Turkish diaspora. **Why sourcing is weak:** His own oral history states clearly that he stayed in Turkey to complete his medical degree (1980-1986) and moved to the U.S. for his wife's scholarship and a voluntary research fellowship. [1] **Best available evidence:** He was a medical resident at Ankara University during the post-coup period, not an exile. [1]

LENS RECOMMENDATION

Lens	Score (0–3)	Justification
Eureka	3	The 1993 discovery of the TNF-α link is a quintessential "paradigm-shifting" moment with high-quality documentation. [2]
Failure	1	While the 1980 coup was a setback, it did not significantly halt his trajectory; he remained in med school and pivoted effectively. [1]
Mentor	2	He speaks highly of his

		father and Spiegelman, but the "pivotal relationship" is less documented than the "pivotal signaling node." [1]
Controversy	2	The PubPeer scrutiny and 2023 corrections provide a balanced, modern tension but do not yet overshadow the primary science. [17]
Human Detail	3	The nomads-to-Harvard journey and the "Dr. G" persona provide a very high level of "texture" and relatability for students. [1]
Last Experiment	2	The 2022 visualization of the liver structure is a perfect "final act" of discovery that moves from molecular to spatial biology. [14]

Primary recommended lens: Eureka (The Discovery of the Hidden Bridge) Why it fits: This lens best supports a narrative intended for BIOL 3030 students. It allows the writer to focus on the 1993 discovery (Scene #1) as the moment when the "wall" between biochemistry and immunology was torn down. It centers the story on a specific biochemical problem (insulin signaling) and how a single discovery changed the textbook definition of a fat cell. **Best opening scene:** Scene #1 (The 1993 lab discovery in the Spiegelman lab). **One quote to use early:** Quote #3 ("The discovery... ushered in a research field"). [10] **Cautions:** The writer must be careful to include the "Complications" (Section E) to avoid a "Great Man" myth, acknowledging that foundational papers have faced modern scrutiny.

Backup lens: Human Detail (The Nomadic Physician) Why it fits: This lens is better if the goal is a more personal, character-driven profile. It uses Scene #3 (The Coup) and Scene #6 (Turkey) to show how his father's rural medical service instilled a sense of purpose that he eventually brought to the Harvard laboratories. **Best opening scene:** Scene #3 (The 1980 coup and the decision to join his wife in the U.S.). **One quote to use early:** Quote #4 (The three people who gave him perspective). [21]

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